

```
evidence/
├── README.md
├── iam/
│   ├── okta-identity-center-federation.png
│   └── aws-temporary-role-access.png
├── authorization/
│   ├── permitted-action.png
│   └── denied-action.png
├── pam/
│   ├── privileged-access-request.png
│   └── jit-access-expiration.png
├── vault/
│   ├── vault-private-healthy.png
│   └── vault-policy-audit.png
├── wazuh/
│   ├── wazuh-private-dashboard.png
│   ├── wazuh-agent-active.png
│   └── wazuh-security-results.png
└── audit-recovery/
    ├── cloudtrail-audit-evidence.png
    └── vault-backup-restore.png
```

iam/

OKTA

**okta**

Admin Console

Dashboard

Directory

People

Groups

Devices

Profile Editor

Directory Integrations

Profile Sources

Search for people, apps and groups

s. asghar consultant-trial-9113435

People

Add person

Reset passwords

More actions

Search for users by first name, primary email or username

Advanced search

Status All Showing 4 of 4

Person & username	Primary email	Status
Charlie Auditor pine85@hotmail.com	pine85@hotmail.com	Active
Bob Developer highroller039@gmail.com	highroller039@gmail.com	Active
Alice Admin marsupialseven@gmail.com	marsupialseven@gmail.com	Password expired
s. asghar s.asghar@consultant.com	s.asghar@consultant.com	Active

FREE TRIAL PLAN

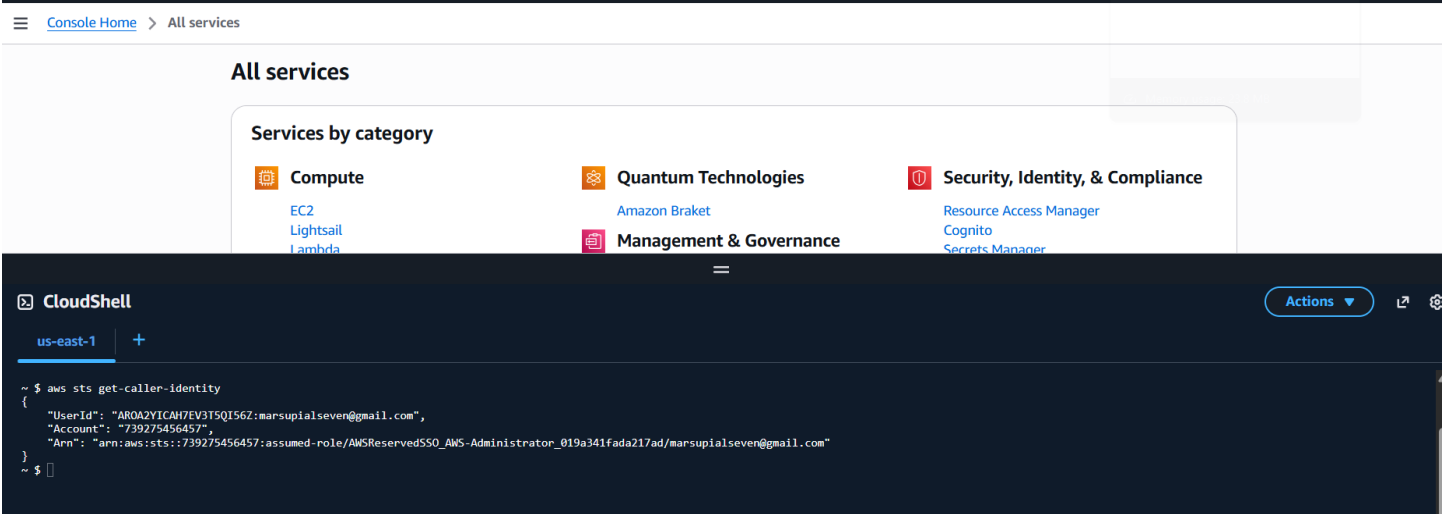
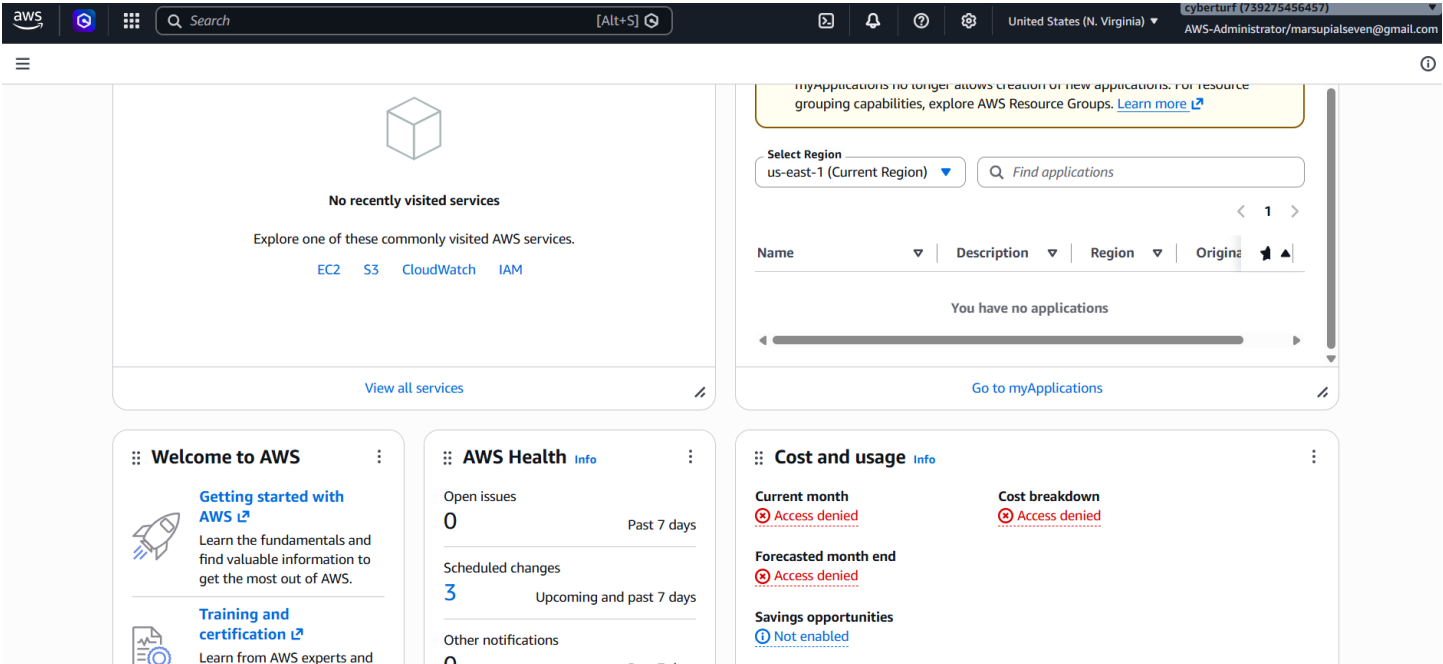
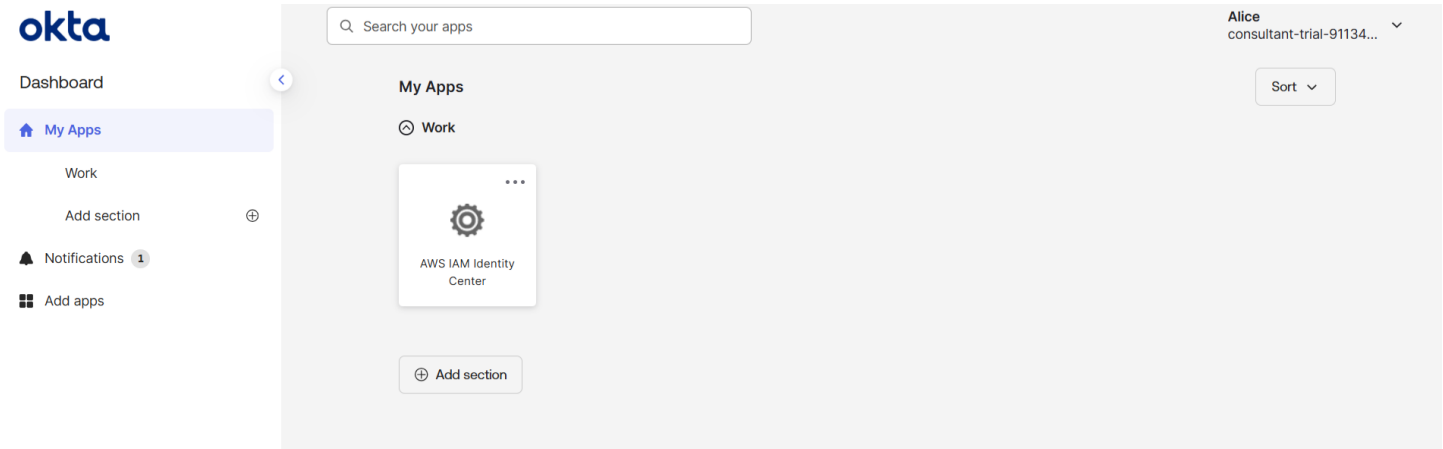
4 of 10 Active users

Updated Aug 4, 2026, 12:27:55 PM

Contact us

/

Alice Cloud Admin Federated via Okta and IAM Identity Center



////////////////////

Bob Developer Federated via Okta and IAM Identity Center



Dashboard

My Apps

Work

Add section

Notifications 1

Add apps

Search your apps

Bob
consultant-trial-91134...

Sort

My Apps

Work



AWS IAM Identity
Center

Add section

Support

Help: s.asghar@consultant.com

aws access portal

Bob

AWS access portal

Accounts

Applications

AWS accounts (1)



Create shortcut

Filter accounts by name, ID, or email address

< 1 > ⚙

Account Name



Account ID

Email



cyberturf

739275456457

marsupialseven@gmail.com

Select your cookie preferences

We use essential cookies and similar tools that are necessary to provide our site and services. We use performance cookies to collect anonymous statistics, so we can understand how customers use our site and make improvements. Essential cookies cannot be deactivated, but you can choose "Customize" or "Decline" to decline performance cookies.

If you agree, AWS and approved third parties will also use cookies to provide useful site features, remember your preferences, and display relevant content, including relevant advertising. To accept or decline all non-essential cookies, choose "Accept" or "Decline." To make more detailed choices, choose "Customize."

Accept

Decline

Customize

aws [Search] [Alt+S] United States (N. Virginia) cyberturf (739275456457) AWS-Developer/highroller039@gmail.com

Console Home info Reset to default layout Add widgets

Recently visited info EC2

Applications (0) info Region: US East (N. Virginia)

myApplications availability change myApplications no longer allows creation of new applications. For resource grouping capabilities, explore AWS Resource Groups. Learn more

CloudShell Actions

us-east-1 +

```
~ $  
~ $ aws sts get-caller-identity  
{  
  "UserId": "AROA2YICAH7E2XOCM3OYJ:highroller039@gmail.com",  
  "Account": "739275456457",  
  "Arn": "arn:aws:sts::739275456457:assumed-role/AWSReservedSSO_AWS-Developer_d3282f041ef38657/highroller039@gmail.com"  
}
```

////////////////////////////////////

authorization/

Bob Developer confederated via Okta and IAM Identity Center (shown above) Successfully Start EC2 Due to Permitted Temporary Access

aws [Search] [Alt+S] United States (N. Virginia) cyberturf (739275456457) AWS-Developer/highroller039@gmail.com

EC2 > Instances

EC2 Dashboard AWS Global View Events

Instances Instances Instance Types Launch Templates Spot Requests

Successfully initiated starting of i-0945b6834101630b8

Instances (1/2) info Connect Instance state Actions Launch instances

Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☒	Bastion-EC2	i-0945b6834101630b8	Pending	t3.micro	-	us-east-1a	-
☐	Private-EC2	i-060e333fcf2fc20c8	Stopped	t3.micro	-	us-east-1a	-

Charlie Auditor confederated via Okta and IAM Identity Center Failed to Start EC2 (Least Privilege)



Dashboard

My Apps

Work

Add section

Notifications 1

Add apps

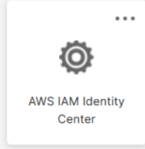
Last sign in: a few seconds ago

© 2026 Okta, Inc.

Search your apps

My Apps

Work



Add section

Support

Help: s.asghar@consultant.com

Request an app

Charlie
consultant-trial-91134...



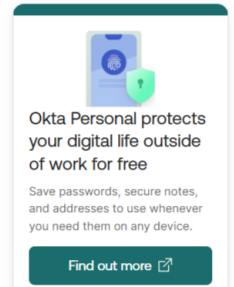
Charlie Auditor
pine85@hotmail.com

Account settings

Recent activity

Display preferences

Sign out



Sign in

ssoins-7223c8ed30cb39fc.portal.us-east-1.app.aws/#/

synthbee

aws access portal

AWS access portal

Accounts

Applications

AWS accounts (1)

Filter accounts by name, ID, or email address

Account Name	Account ID	Email
cyberturf	739275456457	marsupialeven@gmail.com

AWS-Auditor

Access keys

Search [Alt+S]

EC2 Instances

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Failed to start the instance i-0945b6834101630b8
You are not authorized to perform this operation. User: arn:aws:sts::739275456457:assumed-role/AWSReservedSSO_AWS-Auditor_pine85@hotmail.com is not authorized to perform: ec2:StartInstances on resource: arn:aws:ec2:us-east-1:739275456457:instance/i-0945b6834101630b8 because no identity-based policy allows the ec2:StartInstances action. Encoded authorization failure message: xSx-maTrrk6Ppag7tM1m6h3nCVVbcl5bgqZ8oeS9aXsQ-83yTu0Rd8a1G2TFqrkzvT2JTVeS9h4JIPHSR9YiNfoHRzeTdL8ohL-KcS37gIR7o72U-FdutFXXFho4lrWbx-Fg8vRSPbwRdlmDFWPoyvwTcmUy7hwYUGQiz9WTAZLC3POxCi0anDp3ulkXkQ9tDrdcvGMOk9C34iftKGNHoeM09Lttw27z39raLzHrKtwPa-gl0b6cPeGRD3g9lPr6e1S3XYeyXbYNUKRxYDVFQaqDZ8JZKrkcxwxc9-oshQH7bmtYlpax-daC7Atdapc3KpEyFSRfkOw5utOo4dm8AUwXearo530gF5tEYJv6t7pNjelqoZKKdWQ2ZcZmmM6tb106dj7hmm0hTSMOQcJc4bTLQpDjNbNTpz-Ok9ppls1LgDkw86TjCtLKHff8mW5nJav0BvyKav1q-GWU5FFJTW50HZpOfIPaltw6J3paHnaQpx5EHssHLFY7fjMYNZ4PCFGRpT4zm8DCF11mqYG-QieuWUJHpJAxohNxfyVzxtXLux8JedhQzC73h-t-NIDJPVhC5-3y7HPikPlgJLp0RbO2qTEpmDduFYXNfnONV5BzNGPsu6N707Jxm1-AFM3pv3Flam214sng-6PgqRiaQpm7308qThOrsm2j9dbTZzSWt6UsLKNBwVCmxzRoHo-OSqPwDVR0AY3NHq6yy_gjo9lrKciwCw4e8xA-wfcDIAJOrYHMCwNW3PDxHKGK2CkLpDZ7bmlOityMn_rrTWRcDGJzrAVPE9S_BM9SgP4BX4A0ggUW-rja6T0dCYR-4ftUcv99Y-

i-0945b6834101630b8 (Bastion-EC2)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID i-0945b6834101630b8	Public IPv4 address -	Private IPv4 addresses 10.0.0.69
IPv6 address	Instance state	Public DNS

////////////////////////////////

Pam/

Privileged Access Request

The Privileged Access Request is implemented through HashiCorp Vault using a dedicated pam-privileged policy and a protected secret/pam/test-privileged record. The policy grants controlled create, read, update, delete, and list permissions for PAM secrets, while Vault KV versioning maintains a history of changes across four versions. Vault audit logging is enabled through /opt/vault/log/audit.log, providing traceability for privileged access operations and showing whether requests were authorized.

```
[ec2-user@ip-10-20-1-242 pam]$ vault kv get secret/pam/test-privileged
===== Secret Path =====
secret/data/pam/test-privileged

===== Metadata =====
Key          Value
-----
created_time 2026-09-01T18:13:40.235545613Z
custom_metadata <nil>
deletion_time n/a
destroyed     false
version       4

===== Data =====
Key          Value
-----
test_denial  phase6
[ec2-user@ip-10-20-1-242 pam]$
[ec2-user@ip-10-20-1-242 pam]$
[ec2-user@ip-10-20-1-242 pam]$ vault policy read pam-privileged
path "secret/data/pam/*" {
  capabilities = ["create", "read", "update", "delete", "list"]
}

path "secret/metadata/pam/*" {
  capabilities = ["read", "list", "delete"]
}
[ec2-user@ip-10-20-1-242 pam]$
```

/

```
[ec2-user@ip-10-20-1-242 pam]$
[ec2-user@ip-10-20-1-242 pam]$ vault policy read pam-privileged
path "secret/data/pam/*" {
  capabilities = ["create", "read", "update", "delete", "list"]
}

path "secret/metadata/pam/*" {
  capabilities = ["read", "list", "delete"]
}
[ec2-user@ip-10-20-1-242 pam]$
```

/

```
[ec2-user@ip-10-20-1-242 pam]$ vault kv metadata get secret/pam/test-privileged
===== Metadata Path =====
secret/metadata/pam/test-privileged

===== Metadata =====
Key          Value
-----
cas_required  false
created_time  2026-08-19T16:35:28.677555738Z
current_version 4
custom_metadata <nil>
delete_version_after 0s
last_updated_by  map[actor:root client_id:0DHqvq2D77kL2/JTPSZkTMJbkFVmUu0TzMj0jiXcFy8= operation:update]
max_versions    0
oldest_version  0
updated_time    2026-09-01T18:13:40.235545613Z

===== Version 1 =====
Key          Value
-----
created_by    {"actor":"root","client_id":"0DHqvq2D77kL2/JTPSZkTMJbkFVmUu0TzMj0jiXcFy8=","operation":"create"}
created_time  2026-08-19T16:35:28.677555738Z
deleted_by    <nil>
deletion_time n/a
destroyed     false

===== Version 2 =====
Key          Value
-----
created_by    {"actor":"token","client_id":"fh5KDir9gGHZnTGUu00Al0G7X1nNF0r4+HhiEN4GJ8=","operation":"update"}
created_time  2026-08-19T16:52:10.961123852Z
deleted_by    <nil>
deletion_time n/a
destroyed     false
```

```
===== Version 3 =====
Key          Value
-----
created_by    {"actor":"root","client_id":"0DHqvq2D77kL2/JTPSZkTMJbkFVmUu0TzMj0jiXcFy8=","operation":"update"}
created_time  2026-08-25T18:00:16.305778138Z
deleted_by    <nil>
deletion_time n/a
destroyed     false

===== Version 4 =====
Key          Value
-----
created_by    {"actor":"root","client_id":"0DHqvq2D77kL2/JTPSZkTMJbkFVmUu0TzMj0jiXcFy8=","operation":"update"}
created_time  2026-09-01T18:13:40.235545613Z
deleted_by    <nil>
deletion_time n/a
destroyed     false
[ec2-user@ip-10-20-1-242 pam]$
```

```
[ec2-user@ip-10-20-1-242 pam]$ vault audit list -detailed
Path      Type      Description      Replication      Options
-----
file/      file      n/a              replicated        file_path=/opt/vault/log/audit.log
[ec2-user@ip-10-20-1-242 pam]$
```



```
[ec2-user@ip-10-20-1-242 ~]$ sudo grep -i "test-privileged\\secret\\data\\pam" /opt/vault/log/audit.log | tail -20
{"auth":{"accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","display_name":"root","policies":["root"],"policy_results":{"allowed":true},"token_policies":["root"],"token_issue_time":"2026-08-08-19T16:02:55Z","token_type":"service"},"request":{"client_id":"0DHqvg2D77kL2/JTPS2kTMJbkFvMu0TzMi0jiXcFy8=","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","client_token_accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381"},"headers":{"user-agent":["Go-http-client/2.0"]},"id":"106903eb-d28a-c293-2d11-5c95299188df","mount_class":"secret","mount_point":"/sys/","mount_running_version":"v2.0.3+builtin.vault","mount_type":"system","namespace":{"id":"root"},"operation":"read","path":"/sys/internal/ui/mounts/secret/pam/test-privileged","remote_address":"10.20.1.242","remote_port":46394},"time":"2026-09-08T17:03:07.32203743Z","type":"request"}
{"auth":{"accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","display_name":"root","policies":["root"],"policy_results":{"allowed":true},"token_policies":["root"],"token_issue_time":"2026-08-08-19T16:02:55Z","token_type":"service"},"request":{"client_id":"0DHqvg2D77kL2/JTPS2kTMJbkFvMu0TzMi0jiXcFy8=","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","client_token_accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381"},"headers":{"user-agent":["Go-http-client/2.0"]},"id":"106903eb-d28a-c293-2d11-5c95299188df","mount_accessor":"system_c9aeaf7c","mount_class":"secret","mount_point":"/sys/","mount_running_version":"v2.0.3+builtin.vault","mount_type":"system","namespace":{"id":"root"},"operation":"read","path":"/sys/internal/ui/mounts/secret/pam/test-privileged","remote_address":"10.20.1.242","remote_port":46394},"response":{"data":{"accessor":"hmac-sha256:ea0e70929a1f71ff7d25b7af9f97cccb744acd1030b70b6ffdc2f547fa4eca8","config":{"default_lease_ttl":2764800,"force_no_cache":false,"max_lease_ttl":2764800},"deprecation_status":"hmac-sha256:618f309155c74967921b31d7fba07c9a97ac7522817331236804cc0b1ecf542"},"description":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47"},"external_entropy_access":false,"local":false,"options":{"version":"hmac-sha256:5df13fd054455b3d424d81e52d6a1d6655f4fd15d3d31e7f96093ebf17ee27f"},"path":"hmac-sha256:54ee4e561813cdf18600100d258682a0d15d411c33a65b1d6ed0c749163fcd"},"plugin_version":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47"},"running_plugin_version":"hmac-sha256:6d67b1f939d1774ac472ad7e1849caefff11652b1d053fd72d70e1963a1776e"},"running_sha256":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47"},"seal_wrap":false,"type":"hmac-sha256:1527e72f43ab3d33fafb2baa5936cfaf6a8a806d482979c54d2669bae6bdc928"},"uuid":"hmac-sha256:d3b09f21514e6474a062e146ac24bb1aabb6abda50d24645c46925828f14bdf7"},"mount_accessor":"system_c9aeaf7c","mount_class":"secret","mount_point":"/sys/","mount_running_plugin_version":"v2.0.3+builtin.vault","mount_type":"system"},"time":"2026-09-08T17:03:07.322413409Z","type":"response"}
{"auth":{"accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","display_name":"root","policies":["root"],"policy_results":{"allowed":true},"granting_policies":[{"type":"","name":"root","namespace_id":"root","type":"acl"}],"token_policies":["root"],"token_issue_time":"2026-08-19T16:02:55Z","token_type":"service"},"request":{"client_id":"0DHqvg2D77kL2/JTPS2kTMJbkFvMu0TzMi0jiXcFy8=","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","client_token_accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381"},"headers":{"user-agent":["Go-http-client/2.0"]},"id":"47f37ce8-f37d-5191-78d5-636127f916b0","mount_class":"secret","mount_point":"secret/","mount_running_version":"v0.26.2+builtin","mount_type":"kv","namespace":{"id":"root"},"operation":"read","path":"secret/metadata/pam/test-privileged","remote_address":"10.20.1.242","remote_port":46394},"time":"2026-09-08T17:03:07.323637542Z","type":"request"}
{"auth":{"accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","display_name":"root","policies":["root"],"policy_results":{"allowed":true},"granting_policies":[{"type":"","name":"root","namespace_id":"root","type":"acl"}],"token_policies":["root"],"token_issue_time":"2026-08-19T16:02:55Z","token_type":"service"},"request":{"client_id":"0DHqvg2D77kL2/JTPS2kTMJbkFvMu0TzMi0jiXcFy8=","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","client_token_accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381"},"headers":{"user-agent":["Go-http-client/2.0"]},"id":"47f37ce8-f37d-5191-78d5-636127f916b0","mount_accessor":"kv_759923bc","mount_class":"secret","mount_point":"secret/","mount_running_version":"v0.26.2+builtin","mount_type":"kv","namespace":{"id":"root"},"operation":"read","path":"secret/metadata/pam/test-privileged","remote_address":"10.20.1.242","remote_port":46394},"response":{"data":{"cas_req
```

/

```
[ec2-user@ip-10-20-1-242 ~]$ sudo grep 'secret/metadata/pam/test-privileged' /opt/vault/log/audit.log | grep -E '"operation"|"path"|"policy_results"|"time"
{"auth":{"accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","display_name":"root","policies":["root"],"policy_results":{"allowed":true},"granting_policies":[{"type":"","name":"root","namespace_id":"root","type":"acl"}],"token_policies":["root"],"token_issue_time":"2026-08-19T16:02:55Z","token_type":"service"},"request":{"client_id":"0DHqvg2D77kL2/JTPS2kTMJbkFvMu0TzMi0jiXcFy8=","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","client_token_accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381"},"headers":{"user-agent":["Go-http-client/2.0"]},"id":"bd8bcc7c-5b92-5bd2-e759-b04c234451b1","mount_class":"secret","mount_point":"secret/","mount_running_version":"v0.26.2+builtin","mount_type":"kv","namespace":{"id":"root"},"operation":"read","path":"secret/metadata/pam/test-privileged","remote_address":"10.20.1.242","remote_port":35382},"time":"2026-08-24T18:59:05.326653576Z","type":"request"}
{"auth":{"accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","display_name":"root","policies":["root"],"policy_results":{"allowed":true},"granting_policies":[{"type":"","name":"root","namespace_id":"root","type":"acl"}],"token_policies":["root"],"token_issue_time":"2026-08-19T16:02:55Z","token_type":"service"},"request":{"client_id":"0DHqvg2D77kL2/JTPS2kTMJbkFvMu0TzMi0jiXcFy8=","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","client_token_accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381"},"headers":{"user-agent":["Go-http-client/2.0"]},"id":"bd8bcc7c-5b92-5bd2-e759-b04c234451b1","mount_accessor":"kv_759923bc","mount_class":"secret","mount_point":"secret/","mount_running_version":"v0.26.2+builtin","mount_type":"kv","namespace":{"id":"root"},"operation":"read","path":"secret/metadata/pam/test-privileged","remote_address":"10.20.1.242","remote_port":35382},"response":{"data":{"cas_req_uired":false,"created_time":"hmac-sha256:4a492a15fe65b03a2dc127fec25c9c380accf7f797aa1266151de80516cb4ad8","current_version":2,"custom_metadata":null,"delete_version_after":"hmac-sha256:9483c966741c76bca58ef2ca663e40577bb2e3aa86f36ea41f47c97deee2134","last_updated_by":{"actor":"hmac-sha256:a337b49e276b614ffba28be414ca221702878ac939d278bd38267001fe7fbdb"},"operation":"hmac-sha256:36c8a3e43c90c9cd26a301c6cb2412717986f6e845aae53b11d6d671d32f83a"},"entity_id":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47"},"client_id":"hmac-sha256:05d7225effb698ed53cc9d16624ee7bf6d97a83f6eb470fa2c874c6e36e8eb5b"},"max_versions":0,"oldest_version":0,"updated_time":"hmac-sha256:7a8d0fb17c6adbe7f34985190ce49b68b1faa49d334efc3598b58546f05418e0"},"versions":{"1":{"created_by":{"actor":"hmac-sha256:1b9a5e6fde8b29ee23f8000b97c0b3d93a4544c2573bfd2473f27711aa6ad17"},"operation":"hmac-sha256:92999b351af29a85c61573b392bcf14235eaff1f21f3398df8cb092d6840f75f3"},"entity_id":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47"},"client_id":"hmac-sha256:2dfd84cd1be99623f5c2b18bb91736b25575d543e4ad6a50d6dbb5ebfe7cab6"},"created_time":"hmac-sha256:4a492a15fe65b03a2dc127fec25c9c380accf7f797aa1266151de80516cb4ad8","deleted_by":null,"deletion_time":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47"},"destroyed":false},"2":{"created_by":{"actor":"hmac-sha256:a337b49e276b614ffba28be414ca221702878ac939d278bd38267001fe7fbdb"},"operation":"hmac-sha256:36c8a3e43c90c9cd26a301c6cb2412717986f6e845aae53b11d6d671d32f83a"},"entity_id":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47"},"client_id":"hmac-sha256:05d7225effb698ed53cc9d16624ee7bf6d97a83f6eb470fa2c874c6e36e8eb5b"},"created_time":"hmac-sha256:7a8d0fb17c6adbe7f34985190ce49b68b1faa49d334efc3598b58546f05418e0"},"deleted_by":null,"deletion_time":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47"},"destroyed":false},"mount_accessor":"kv_759923bc","mount_class":"secret","mount_point":"secret/","mount_running_plugin_version":"v0.26.2+builtin","mount_type":"kv"},"time":"2026-08-24T18:59:05.32723554Z","type":"response"}
{"auth":{"accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","display_name":"root","policies":["root"],"policy_results":{"allowed":true},"granting_policies":[{"type":"","name":"root","namespace_id":"root","type":"acl"}],"token_policies":["root"],"token_issue_time":"2026-08-19T16:02:55Z","token_type":"service"},"request":{"client_id":"0DHqvg2D77kL2/JTPS2kTMJbkFvMu0TzMi0jiXcFy8=","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","client_token_accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381"},"headers":{"user-agent":["Go-http-client/2.0"]},"id":"00b2d0aa-79fe-c369-7fc1-125c294ce5de","mount_class":"secret","mount_point":"secret/","mount_running_version":"v0.26.2+builtin","mount_type":"kv","namespace":{"id":"root"},"operation":"read","path":"secret/metadata/pam/test-privileged","remote_address":"10.20.1.242","remote_port":58416},"time":"2026-08-25T17:58:29.492710458Z","type":"request"}
{"auth":{"accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","display_name":"root","policies":["root"],"policy_results":{"allowed":true},"granting_policies":[{"type":"","name":"root","n
```

JIT TTL values are recorded

JIT / Time-Bound Privileged Access — Vault Audit Evidence

This evidence shows a pam-test-privileged Vault token issued with a **600-second (10-minute) TTL**.

The temporary token grants privileged PAM access only for the defined lifetime. This demonstrates **JIT/time-bound privileged access with auditable token issuance**.


```
[ec2-user@ip-10-20-1-242 ~]$ sudo grep -F "pam-user" /opt/vault/log/audit.log | tail -n 10 | sed -E 's/"/(client_token|accessor|client_token_accessor|client_id):"/'
{"auth":{"accessor":"[REDACTED]","client_token":"[REDACTED]","display_name":"token","policies":["default","pam-user"],"policy_results":{"allowed":true,"granting_policies":[{"type":"","name":"default","namespace_id":"root","type":"acl"}]},"token_policies":["default","pam-user"],"token_issue_time":"2026-08-25T17:04:04Z","token_ttl":30,"token_type":"service"},"request":{"client_id":"[REDACTED]","client_token":"[REDACTED]","client_token_accessor":"[REDACTED]","headers":{"user-agent":["Go-http-client/2.0"]},"id":"5a0d71e1-9ef7-50e6-13f1-699cac5a54e5","mount_class":"auth","mount_point":"auth/token/","mount_running_version":"v2.0.3+builtin.vault","mount_type":"token","namespace":{"id":"root"},"operation":"read","path":"auth/token/lookup-self","remote_address":"10.20.1.242","remote_port":40092},"time":"2026-08-25T17:04:34.123645275Z"},"type":"request"}
{"auth":{"accessor":"[REDACTED]","client_token":"[REDACTED]","display_name":"token","policies":["default","pam-user"],"policy_results":{"allowed":true,"granting_policies":[{"type":"","name":"default","namespace_id":"root","type":"acl"}]},"token_policies":["default","pam-user"],"token_issue_time":"2026-08-25T17:04:04Z","token_ttl":30,"token_type":"service"},"request":{"client_id":"[REDACTED]","client_token":"[REDACTED]","client_token_accessor":"[REDACTED]","headers":{"user-agent":["Go-http-client/2.0"]},"id":"5a0d71e1-9ef7-50e6-13f1-699cac5a54e5","mount_accessor":"auth_token_46c461b4","mount_class":"auth","mount_point":"auth/token/","mount_running_version":"v2.0.3+builtin.vault","mount_type":"token","namespace":{"id":"root"},"operation":"read","path":"auth/token/lookup-self","remote_address":"10.20.1.242","remote_port":40092},"response":{"data":{"accessor":"[REDACTED]","creation_time":1787677444,"creation_ttl":30,"display_name":"hmac-sha256:a337b49e276b614ffba28be414ca221702878aca939d278bd38267001fe7fbbd","entity_id":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47","expire_time":"2026-08-25T17:04:34.857857936Z","explicit_max_ttl":0,"id":"hmac-sha256:615e64eb09e6713fdd5ee37a7293fc20f97c518b7047a2b31acebbcf4307bdc663","issue_time":"2026-08-25T17:04:04.857864262Z","meta":null,"num_users":0,"orphan":false,"path":"hmac-sha256:6f4eb9d742fb6ae438b460df7b6c625e20b3d055912f12205bfbfbef186044cb","policies":["hmac-sha256:f93ea42cbfc535c19e07b5d7495d434cf67f28fcd174e973f83508066c20dcf","hmac-sha256:bcf2afba5c77e0590f136a35e5bda0be2fc195514fd10972e9f0b451b3b20d5"],"renewable":true,"ttl":0,"type":"hmac-sha256:9a7312f78e565cab496eb6a4bc994311917dcecbdbe54293aef4c4b5996c5907"},"mount_accessor":"auth_token_46c461b4","mount_class":"auth","mount_point":"auth/token/","mount_running_plugin_version":"v2.0.3+builtin.vault","mount_type":"token"},"time":"2026-08-25T17:04:34.123971269Z"},"type":"response"}
{"auth":{"accessor":"[REDACTED]","client_token":"[REDACTED]","display_name":"root","policies":["root"],"policy_results":{"allowed":true,"granting_policies":[{"type":"","name":"root","namespace_id":"root","type":"acl"}]},"token_policies":["root"],"token_issue_time":"2026-08-19T16:02:55Z","token_type":"service"},"request":{"client_id":"[REDACTED]","client_token":"[REDACTED]","client_token_accessor":"[REDACTED]","data":{"display_name":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47","entity_alias":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47","explicit_max_ttl":"hmac-sha256:9483c966741cc76bca58ef2ca663e40577bb2e3aa86f36ea41f47c97deee2134","num_users":"hmac-sha256:63490fc3e7f99933ce4b9e52fe8c31de0b90dfb6d91da52673934bb59302103d","period":"hmac-sha256:9483c966741cc76bca58ef2ca663e40577bb2e3aa86f36ea41f47c97deee2134"},"policies":["hmac-sha256:bcf2afba5c77e0590f136a35e5bda0be2fc195514fd10972e9f0b451b3b20d5"],"renewable":true,"ttl":"hmac-sha256:408b93d5b26e79a8368b60f188e9491a7fc2876ab6f273ff9093176d22743b2b"},"type":"hmac-sha256:9a7312f78e565cab496eb6a4bc994311917dcecbdbe54293aef4c4b5996c5907"},"headers":{"user-agent":["Go-
```

```
[ec2-user@ip-10-20-1-242 ~]$ sudo grep -RniE 'password|secret|token|access_key|private_key' \
/etc/vault.d /opt/vault/log 2>/dev/null | head -n 30 | grep ttl
/opt/vault/log/audit.log:5:{"auth":{"accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","display_name":"root","policies":["root"],"policy_results":{"allowed":true,"granting_policies":[{"type":"","name":"root","namespace_id":"root","type":"acl"}]},"token_policies":["root"],"token_issue_time":"2026-08-19T16:02:55Z","token_type":"service"},"request":{"client_id":"0DHQvq2D77kL2/JTPS2kTMJbKfVmuU0TzMi0jiXcFY8","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","client_token_accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381","data":{"display_name":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47","entity_alias":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47","explicit_max_ttl":"hmac-sha256:9483c966741cc76bca58ef2ca663e40577bb2e3aa86f36ea41f47c97deee2134","num_users":"hmac-sha256:63490fc3e7f99933ce4b9e52fe8c31de0b90dfb6d91da52673934bb59302103d","period":"hmac-sha256:9483c966741cc76bca58ef2ca663e40577bb2e3aa86f36ea41f47c97deee2134"},"policies":["hmac-sha256:da4d718193e6278c5e8e619dbc651ae820e7ffe7dec697ce144f8cc66d61c0"],"renewable":true,"ttl":"hmac-sha256:6040363013a59df3579ffff2006cac8796dccc8f8ac6fbc8e886555b3589a89f0"},"type":"hmac-sha256:9a7312f78e565cab496eb6a4bc994311917dcecbdbe54293aef4c4b5996c5907"},"headers":{"user-agent":["Go-http-client/2.0"]},"id":"86f99974-14a5-5d3a-7e87-b2e7856ca939"},"mount_accessor":"auth_token_46c461b4","mount_class":"auth","mount_point":"auth/token/","mount_running_version":"v2.0.3+builtin.vault","mount_type":"token","namespace":{"id":"root"},"operation":"update","path":"auth/token/create","remote_address":"10.0.0.69","remote_port":44326,"time":"2026-08-19T17:56:16.301993974Z"},"type":"request"}
/opt/vault/log/audit.log:6:{"auth":{"accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","display_name":"root","policies":["root"],"policy_results":{"allowed":true,"granting_policies":[{"type":"","name":"root","namespace_id":"root","type":"acl"}]},"token_policies":["root"],"token_issue_time":"2026-08-19T16:02:55Z","token_type":"service"},"request":{"client_id":"0DHQvq2D77kL2/JTPS2kTMJbKfVmuU0TzMi0jiXcFY8","client_token":"hmac-sha256:b88e12caa21f89880bf0361d084901715a06ce27ad139ead0e382922fbc149f5","client_token_accessor":"hmac-sha256:f232509de151797f9510d5ae22860c256ce0d8071c9f48f5cedbb1f6be49381","data":{"display_name":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47","entity_alias":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47","explicit_max_ttl":"hmac-sha256:9483c966741cc76bca58ef2ca663e40577bb2e3aa86f36ea41f47c97deee2134","num_users":"hmac-sha256:63490fc3e7f99933ce4b9e52fe8c31de0b90dfb6d91da52673934bb59302103d","period":"hmac-sha256:9483c966741cc76bca58ef2ca663e40577bb2e3aa86f36ea41f47c97deee2134"},"policies":["hmac-sha256:da4d718193e6278c5e8e619dbc651ae820e7ffe7dec697ce144f8cc66d61c0"],"renewable":true,"ttl":"hmac-sha256:6040363013a59df3579ffff2006cac8796dccc8f8ac6fbc8e886555b3589a89f0"},"type":"hmac-sha256:9a7312f78e565cab496eb6a4bc994311917dcecbdbe54293aef4c4b5996c5907"},"headers":{"user-agent":["Go-http-client/2.0"]},"id":"86f99974-14a5-5d3a-7e87-b2e7856ca939"},"mount_accessor":"auth_token_46c461b4","mount_class":"auth","mount_point":"auth/token/","mount_running_version":"v2.0.3+builtin.vault","mount_type":"token","namespace":{"id":"root"},"operation":"update","path":"auth/token/create","remote_address":"10.0.0.69","remote_port":44326,"response":{"auth":{"accessor":"hmac-sha256:f39ceaa83a439c6ab0da773ea0c69f737d798bbfd71dbb5d79c3686037cd6de","client_token":"hmac-sha256:64bfe18b41629cc331142bb78be7cec0fbeb5b4ab0112dcdb7d2e6abd4925ccf"},"display_name":"token","policies":["default","pam-privileged"],"token_policies":["default","pam-privileged"],"token_type":"service"},"token_accessor":"auth_token_46c461b4","mount_class":"auth","mount_point":"auth/token/","mount_running_plugin_version":"v2.0.3+builtin.vault","mount_type":"token"},"time":"2026-08-19T17:56:16.320080609Z"},"type":"response"}
/opt/vault/log/audit.log:9:{"auth":{"accessor":"hmac-sha256:f39ceaa83a439c6ab0da773ea0c69f737d798bbfd71dbb5d79c3686037cd6de","client_token":"hmac-sha256:64bfe18b41629cc331142bb78be7cec0fbeb5b4ab0112dcdb7d2e6abd4925ccf","display_name":"token","policies":["default","pam-privileged"],"policy_results":{"allowed":true},"token_policies":["default","pam-privileged"],"token_issue_time":"2026-08-19T17:56:16Z","token_ttl":600,"token_type":"service"},"request":{"client_id":"fh5K0Dir9gGHZnTGUU0A10G7X1nNF00r4+HhiEN4GJ8","client_token":"hmac-sha256:543503a1fe0721b1cddb07559ae77fee02a4a59436789f010d6610341cd6df103","client_token_accessor":"hmac-sha256:f39ceaa83a439c6ab0da773ea0c69f737d798bbfd71dbb5d79c3686037cd6de"},"headers":{"user-agent":["Go-http-client/2.0"]},"id":"37f88b60-7f93-7fd7-0043-ae80e
```

```
[ec2-user@ip-10-20-1-242 ~]$ sudo grep 'token_ttl:600' /opt/vault/log/audit.log | tail -1
{"auth":{"accessor":"hmac-sha256:eb276958b3c85a60f1f2bbf23169e1451b01091e8881757c5f7e576cfc49261","client_token":"hmac-sha256:d815bcbf615268ad4365ada3825bb4313521b16034a1c12be394f4ce317fc1c2","display_name":"token","policies":["default","pam-user"],"policy_results":{"allowed":true,"granting_policies":[{"type":"","name":"default","namespace_id":"root","type":"acl"}]},"token_policies":["default","pam-user"],"token_issue_time":"2026-09-01T18:22:47Z","token_ttl":600,"token_type":"service"},"request":{"client_id":"CKBJUZeFH00u33zfkfzj9QROAhuyx9JYVKPenwXfQ","client_token":"hmac-sha256:d2f3ff820fd14d59c7b3ee964ca9a6f5669da789fc1fa69e90e19442f53df26e","client_token_accessor":"hmac-sha256:eb276958b3c85a60f1f2bbf23169e1451b01091e8881757c5f7e576cfc49261","headers":{"user-agent":["Go-http-client/2.0"]},"id":"f73fade3-1f7b-845d-ec8a-472f7e8ff8a2","mount_accessor":"auth_token_46c461b4","mount_class":"auth","mount_point":"auth/token/","mount_running_version":"v2.0.3+builtin.vault","mount_type":"token","namespace":{"id":"root"},"operation":"read","path":"auth/token/lookup-self","remote_address":"10.20.1.242","remote_port":60004},"response":{"data":{"accessor":"hmac-sha256:eb276958b3c85a60f1f2bbf23169e1451b01091e8881757c5f7e576cfc49261","creation_time":1788286967,"creation_ttl":600,"display_name":"hmac-sha256:a337b49e276b614ffba28be414ca221702878aca939d278bd38267001fe7fbbd","entity_id":"hmac-sha256:32405e2855435c703f2651cb135aff50beacc8434688b4620d7541666839ea47","expire_time":"2026-09-01T18:32:47.395278408Z","explicit_max_ttl":0,"id":"hmac-sha256:d815bcbf615268ad4365ada3825bb4313521b16034a1c12be394f4ce317fc1c2","issue_time":"2026-09-01T18:22:47.395283906Z","meta":null,"num_users":0,"orphan":false,"path":"hmac-sha256:6f4eb9d742fb6ae438b460df7b6c625e20b3d055912f12205bfbfbef186044cb"},"policies":["hmac-sha256:8f93ea42cbfc535c19e07b5d7495d434cf67f28fcd174e973f83508066c20dcf","hmac-sha256:bcf2afba5c77e0590f136a35e5bda0be2fc195514fd10972e9f0b451b3b20d5"],"renewable":true,"ttl":282,"type":"hmac-sha256:9a7312f78e565cab496eb6a4bc994311917dcecbdbe54293aef4c4b5996c5907"},"mount_accessor":"auth_token_46c461b4","mount_class":"auth","mount_point":"auth/token/","mount_running_plugin_version":"v2.0.3+builtin.vault","mount_type":"token"},"time":"2026-09-01T18:28:05.184469733Z"},"type":"response"}
[ec2-user@ip-10-20-1-242 ~]$
```


////////////////////

Vault/

Vault health

```
[ec2-user@ip-10-20-1-242 ~]$ curl -k -sS https://10.20.1.242:8200/v1/sys/health
{"initialized":true,"sealed":false,"standby":false,"performance_standby":false,"replication_performance_mode":"disabled","replication_dr_mode":"disabled","server_time_utc":1788199985,"version":"2.0.3","enterprise":false,"cluster_name":"vault-cluster-835dd1f3","cluster_id":"9b088747-2339-fc8c-a4c8-0772d03e1be1","echo_duration_ms":0,"clock_skew_ms":0,"replication_primary_canary_age_ms":0,"removed_from_cluster":false}
[ec2-user@ip-10-20-1-242 ~]$
```

Vault Policy Audit

The pam-privileged policy grants privileged users full CRUD/list access to PAM secrets under secret/data/pam/*. It also allows read/list/delete access to the corresponding secret metadata. This screenshot provides evidence of Vault policy-based privileged access control.

```
[ec2-user@ip-10-20-1-242 ~]$
echo "=== VAULT POLICY AUDIT ==="
echo "— Policies —"
vault policy list
echo
echo "— pam-privileged policy —"
vault policy read pam-privileged
=== VAULT POLICY AUDIT ===
— Policies —
default
default-ceiling
pam-privileged
pam-user
root

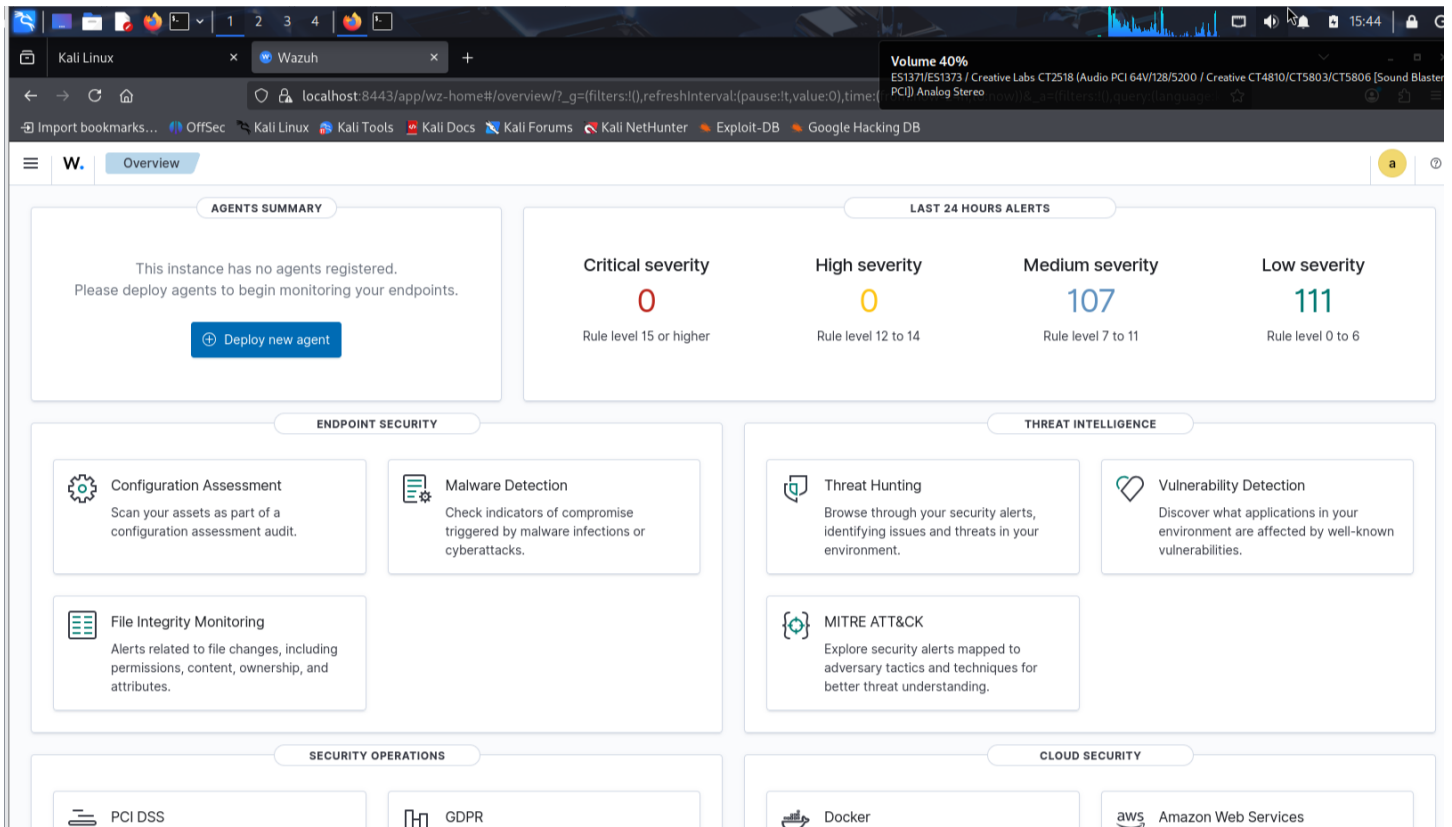
— pam-privileged policy —
path "secret/data/pam/*" {
  capabilities = ["create", "read", "update", "delete", "list"]
}

path "secret/metadata/pam/*" {
  capabilities = ["read", "list", "delete"]
}
[ec2-user@ip-10-20-1-242 ~]$
```

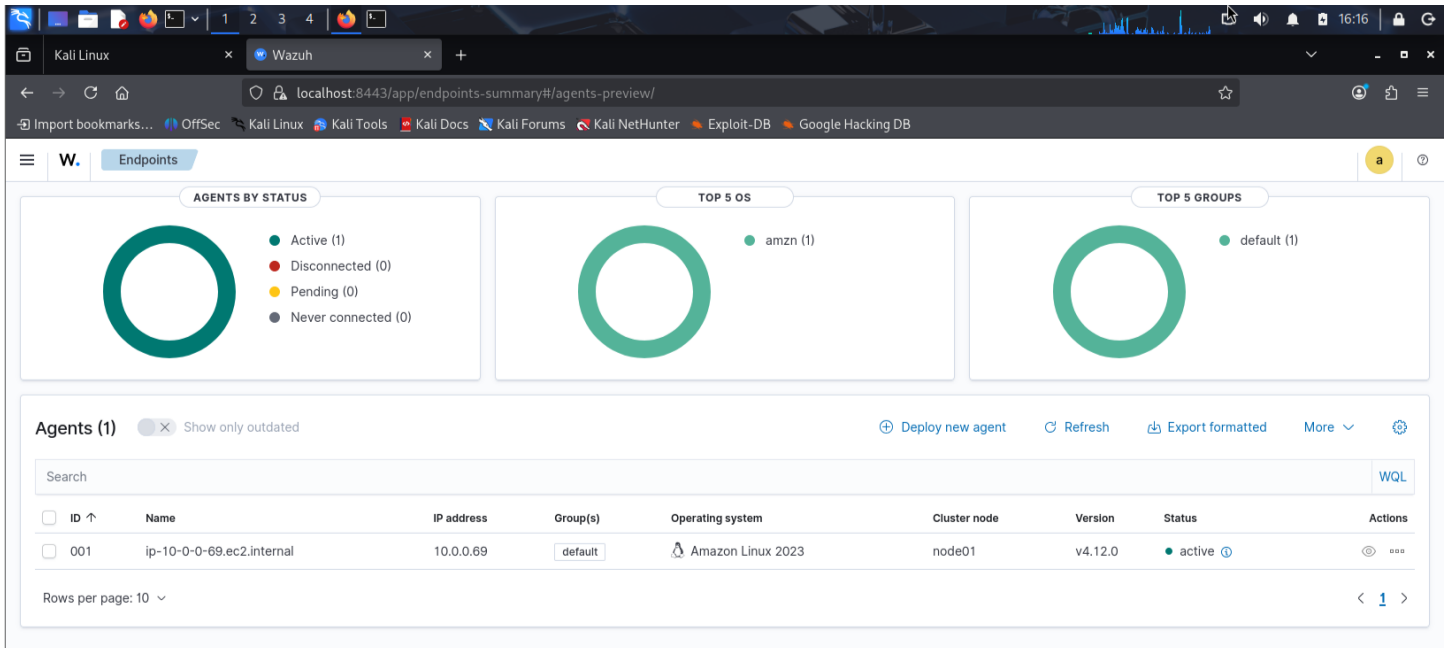
////////////////////

Wazuh/

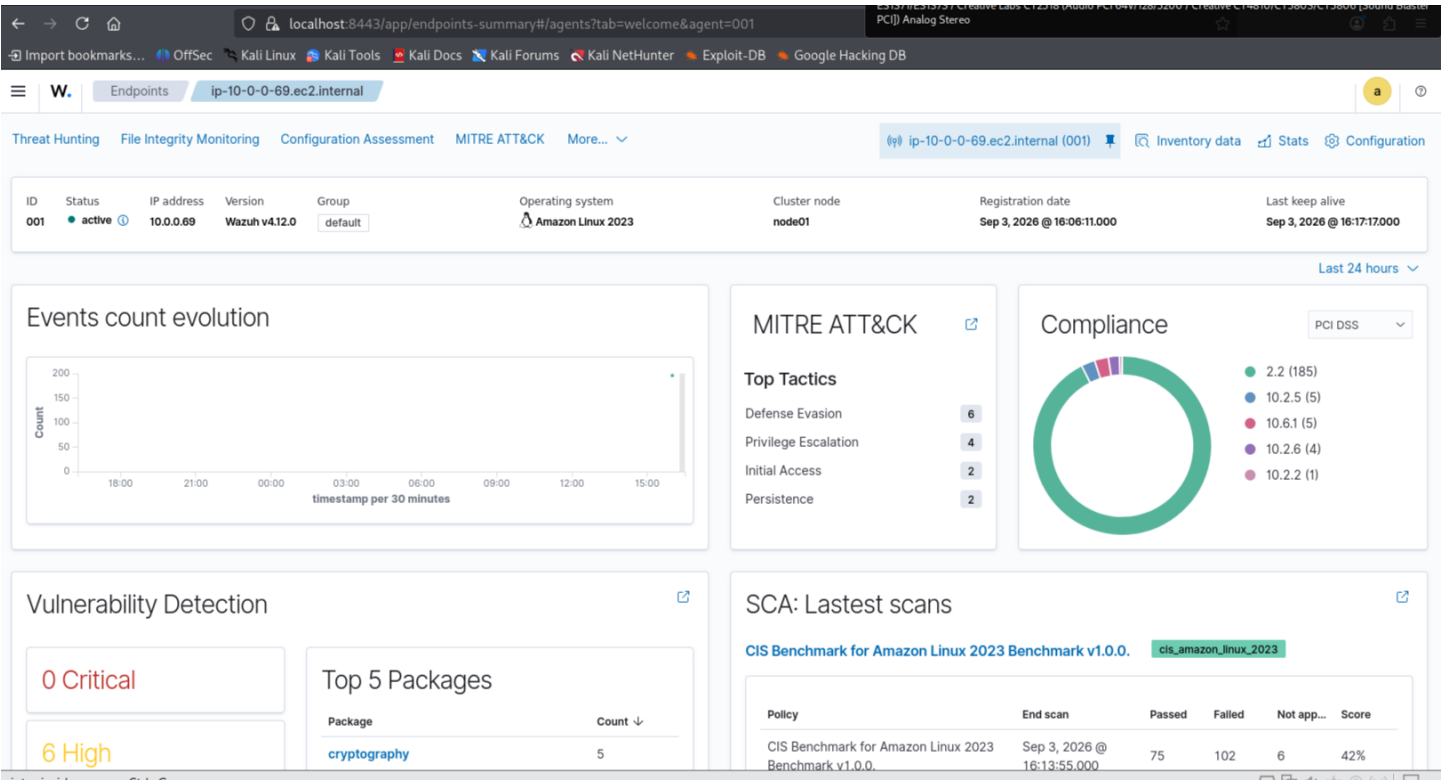
Private Dash Board



Wazuh Agent Active



Wazuh Security Results AS IS



////////////////////

Audit-Recovery/

Vault Cloudtrail Audit Evidence

AWS CloudTrail Audit Evidence — Vault EC2 Recovery Environment
CloudTrail records EC2 lifecycle and IAM role activity associated with the Vault recovery environment on September 7, 2026.

aws

Search

[Alt+S]

Ask Amazon Q

United States (N. Virginia)

EC2

Instances

i-0295c531f397dcbe2

EC2

Dashboard

Instance summary for i-0295c531f397dcbe2 (pam-vault-01)

Updated 13 minutes ago

Connect

Instance state

CloudShell

us-east-1

```
~ $
~ $ clear
~ $ aws cloudtrail lookup-events \
> --region us-east-1 \
> --lookup-attributes AttributeKey=ResourceName,AttributeValue=i-0295c531f397dcbe2 \
> --start-time "2026-09-07T00:00:00Z" \
> --end-time "2026-09-08T00:00:00Z" \
> --max-results 50 \
> --query 'Events[].[Time:EventTime,Event:EventName,User:Username,Source:EventSource]' \
> --output table
```

LookupEvents			
Event	Source	Time	User
StopInstances	ec2.amazonaws.com	2026-09-07T19:59:39+00:00	root
AssumeRole	sts.amazonaws.com	2026-09-07T19:45:20+00:00	None
AssumeRole	sts.amazonaws.com	2026-09-07T19:45:20+00:00	None
AssumeRole	sts.amazonaws.com	2026-09-07T19:08:48+00:00	None
AssumeRole	sts.amazonaws.com	2026-09-07T19:08:48+00:00	None
StartInstances	ec2.amazonaws.com	2026-09-07T19:08:45+00:00	root

Vault backup/restore

Created and retained a Raft snapshot, restored the September 2 backup, and verified post-restore Vault availability, Raft leadership, and recovery of the PAM KV record secret/pam/test-privileged. **“Post-Restore Verification”**

```
[ec2-user@ip-10-20-1-242 ~]$ vault status
vault operator raft list-peers
vault secrets list
Key                               Value
Seal Type                         shamir
Initialized                       true
Sealed                           false
Total Shares                     5
Threshold                        3
Version                          2.0.3
Build Date                       2026-06-17T12:39:45Z
Storage Type                     raft
Cluster Name                     vault-cluster-835dd1f3
Cluster ID                       9b088747-2339-fc8c-a4c8-0772d03e1be1
Removed From Cluster             false
HA Enabled                       true
HA Cluster                       https://10.20.1.242:8201
HA Mode                          active
Active Since                     2026-09-07T19:37:51.189079188Z
Raft Committed Index             1475
Raft Applied Index               1475
Node      Address                State      Voter
vault-01  10.20.1.242:8201             leader     true
Path      Type                          Accessor      Description
agent-registry/  agent_registry              agent-registry_f9a79505  agent registry
cubbyhole/      cubbyhole                   cubbyhole_22fb1ca3      per-token private secret storage
identity/       identity                    identity_9d529b5b       identity store
secret/         kv                          kv_759923bc            n/a
sys/            system                      system_c9aeaf7c         system endpoints used for control, policy and debugging
[ec2-user@ip-10-20-1-242 ~]$
```

```
[ec2-user@ip-10-20-1-242 ~]$ vault status
Key      Value
-----
Seal Type      shamir
Initialized    true
Sealed         false
Total Shares   5
Threshold      3
Version        2.0.3
Build Date     2026-06-17T12:39:45Z
Storage Type    raft
Cluster Name    vault-cluster-835dd1f3
Cluster ID      9b088747-2339-fc8c-a4c8-0772d03e1be1
Removed From Cluster false
HA Enabled      true
HA Cluster      https://10.20.1.242:8201
HA Mode         active
Active Since    2026-09-07T19:27:58.56943863Z
Raft Committed Index 1449
Raft Applied Index 1449
[ec2-user@ip-10-20-1-242 ~]$
[ec2-user@ip-10-20-1-242 ~]$
[ec2-user@ip-10-20-1-242 ~]$ vault operator raft list-peers
Node      Address      State      Voter
-----
vault-01  10.20.1.242:8201 leader    true
[ec2-user@ip-10-20-1-242 ~]$
[ec2-user@ip-10-20-1-242 ~]$
[ec2-user@ip-10-20-1-242 ~]$ ls -lh /opt/vault/backup/vault-raft-20260902-155402.snap
-rw-r--r-- 1 ec2-user ec2-user 26K Sep  2 15:54 /opt/vault/backup/vault-raft-20260902-155402.snap
[ec2-user@ip-10-20-1-242 ~]$
```

/

```
[ec2-user@ip-10-20-1-242 ~]$
[ec2-user@ip-10-20-1-242 ~]$ vault operator raft snapshot restore /opt/vault/backup/vault-raft-20260902-155402.snap
[ec2-user@ip-10-20-1-242 ~]$
[ec2-user@ip-10-20-1-242 ~]$
```

/

```
[ec2-user@ip-10-20-1-242 ~]$ ls -lh /opt/vault/backup/vault-raft-20260902-155402.snap
-rw-r--r-- 1 ec2-user ec2-user 26K Sep  2 15:54 /opt/vault/backup/vault-raft-20260902-155402.snap
[ec2-user@ip-10-20-1-242 ~]$
[ec2-user@ip-10-20-1-242 ~]$
[ec2-user@ip-10-20-1-242 ~]$ vault kv list secret/
Keys
-----
pam/
[ec2-user@ip-10-20-1-242 ~]$
[ec2-user@ip-10-20-1-242 ~]$ vault kv list secret/pam/
Keys
-----
test-privileged
[ec2-user@ip-10-20-1-242 ~]$
[ec2-user@ip-10-20-1-242 ~]$ vault kv get secret/pam/test-privileged
===== Secret Path =====
secret/data/pam/test-privileged

===== Metadata =====
Key      Value
-----
created_time 2026-09-01T18:13:40.235545613Z
custom_metadata <nil>
deletion_time n/a
destroyed      false
version        4

===== Data =====
Key      Value
-----
test_denial phase6
[ec2-user@ip-10-20-1-242 ~]$
[ec2-user@ip-10-20-1-242 ~]$ history | grep "raft snapshot restore"
351 vault operator raft snapshot restore -help
352 vault operator raft snapshot restore -help |grep -i snapshot
353 vault operator raft snapshot restore -output-policy
354 vault operator raft snapshot restore -output-policy /opt/vault/backup/vault-raft-20260902-155402.snap
```

/

```
ec2-user@ip-10-20-1-242:~
Session Actions Edit View Help
=== POST-RESTORE VAULT VERIFICATION ===
Key Value
Seal Type shamir
Initialized true
Sealed false
Total Shares 5
Threshold 3
Version 2.0.3
Build Date 2026-06-17T12:39:45Z
Storage Type raft
Cluster Name vault-cluster-835dd1f3
Cluster ID 9b088747-2339-fc8c-a4c8-0772d03e1be1
Removed From Cluster false
HA Enabled true
HA Cluster https://10.20.1.242:8201
HA Mode active
Active Since 2026-09-08T15:05:23.641011688Z
Raft Committed Index 1537
Raft Applied Index 1537
=== RAFT LEADER ===
Node Address State Voter
vault-01 10.20.1.242:8201 leader true
=== RESTORED PAM RECORD ===
Keys
test-privileged
===== Secret Path =====
secret/data/pam/test-privileged
===== Metadata =====
Key Value
created_time 2026-09-01T18:13:40.235545613Z
custom_metadata <nil>
deletion_time n/a
destroyed false
version 4
```

```
===== Metadata =====
Key Value
created_time 2026-09-01T18:13:40.235545613Z
custom_metadata <nil>
deletion_time n/a
destroyed false
version 4
===== Data =====
Key Value
test_denial phase6
[ec2-user@ip-10-20-1-242 ~]$
```

////////////////////

Jira Flow

For you

Recent

Starred

Apps

Plans

Spaces

Recent

Security_Operations_Workflow_ka...

Change_Management

More spaces

Recommended

Collect IT requests

Filters

Dashboards

Teams

Goals

Projects

Customize sidebar

Search

Create

2 days left

1

PAM Privileged Access Request - Production Database Read Access

Description

PAM Privileged Access Request - Production Database Read Access

PAM REQUEST

Request ID: SOWK-2

Requester: s. asghar

Target: Production database

Privilege: Read-only privileged access

Business justification:

Temporary access required to investigate an authorized database issue.

Requested duration:

30 minutes

Approver:

System/business owner

Approval timestamp:

Pending

Details

Assignee

Unassigned

Assign to me

Labels

pam-request

Parent

None

Team

None

Due date

None

Start date

None

Priority

Medium

Reporter

s. asghar

Development

For you

Recent

Starred

Apps

Plans

Spaces

Recent

Security_Operations_Workflow_ka...

Change_Management

More spaces

Recommended

Collect IT requests

Filters

Dashboards

Teams

Goals

Projects

Customize sidebar

Search

Create

2 days left

1

SA

Add a comment...

Suggest a reply... Who is working on this...? Status update...

Pro tip: press **M** to comment

SA

s. asghar

now

PAM Authorization: AUTHORIZED

PAM: HashiCorp Vault

Policy: pam-user

Access: Read-only to approved PAM secret path

Duration: 30 minutes

Authorization evidence: Existing Vault TTL/audit validation

Credential handling: No credentials recorded in Jira.

SA

s. asghar

now

PAM Approval Decision: APPROVED

Approver: System/business owner

Decision: Approved for 30 minutes of read-only privileged access.

Details

Assignee

Labels

Parent

Team

Due date

Start date

Priority

Reporter

Development

Automation

Created 1 minute ago

Updated now

Configure